

Atomistic DWBA for Unit Cells in a Graded Interface

orGUI design note

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Abstract

This note specifies the first-order distorted-wave Born approximation needed to add finite atomic models to orGUIT's graded optical reference. It derives, rather than quotes, both the matrix element and the factor that converts it into a measurable amplitude: Born expansion, Green function of the stratified reference built explicitly from two homogeneous solutions and their Wronskian, lattice expansion of the index contrast, and lateral collapse. The Wronskian is shown to equal $2ik_{z,0}$ independently of the prepared stack, which is the origin of every factor $1/k_z$ in this subject; the reciprocal exit field then follows from a boundary-value problem, with no far-field approximation over the illuminated footprint — an approximation that is in fact invalid on the specular rod.

The result is a single matrix element $F_{\mathbf{h}}$ per crystal truncation rod, normalised to one reference lateral cell. A generated `UnitCell`, whether it belongs to a `Film`, an `EpitaxyInterface`, or a surface model, is evaluated in every optical medium that contains its atoms, and contributes to $F_{\mathbf{h}}$ as the coherent sum of the four Renaud incident/exit channels — which are simply the two branches of each reference field. Because the prepared optical reference is laterally uniform, its constant $\delta_m + i\beta_m$ density enters the $\mathbf{h} = \mathbf{0}$ Fourier coefficient and no other: it is part of $F_{\mathbf{0}}$, not something subtracted afterwards from an atomic sum, and it is absent from every genuine rod. That contribution is partitioned among unit cells without changing the profile-level result, so signed occupancies, strain, lateral-area conversion, and optical-profile simplification remain consistent.

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Chapter 1

Reference problem, conventions, and the master formula

1.1 Scope

The unperturbed electric fields and the unperturbed specular reflection amplitude are assumed to be available from the layered-field solver. The problem addressed here is the remaining atomistic matrix element for finite unit cells in the graded interface region. The terminal semi-infinite bulk is included only to show that the new formula reduces to the already implemented bulk result.

The derivation is electromagnetic throughout. Following Renaud, Lazzari, and Leroy, the incident field and the reciprocal field launched from the detector are denoted by electric fields $\mathbf{E}_0$; no quantum-mechanical wavefunction notation is used. The medium is nonmagnetic, isotropic, linear, and reciprocal. Magnetic and tensor-resonant scattering are outside the present scope.

This chapter derives, rather than quotes, both the matrix element and the factor that converts it into a measurable rod amplitude. The chain is: Born expansion, Green function of the stratified reference built from two homogeneous solutions and their Wronskian, lattice expansion of the index contrast, and lateral collapse. No far-field (Fraunhofer) approximation over the illuminated footprint is used at any point; Sec. 1.5.5 explains why one must not be.

1.2 Fourier, surface, and refractive-index conventions

We use the crystallographic space-time convention

$$\mathbf{E}(\mathbf{r}, t) \propto \exp[i(\omega t - \mathbf{k} \cdot \mathbf{r})], \quad k_0 = \frac{2\pi}{\lambda}, \quad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i, \quad (1.1)$$

and the Fourier transform

$$\tilde{g}(\mathbf{Q}) = \int g(\mathbf{r}) \exp(i\mathbf{Q} \cdot \mathbf{r}) \, d^3r. \quad (1.2)$$

The z axis is the outward surface normal: vacuum is at large positive z and the bulk extends toward negative z . Momentum transfer is in $\text{\AA}^{-1}$ and all real-space coordinates below are in $\text{\AA}$.

The prepared optical reference consists of planar media $m = 0, \dots, M$. In medium m ,

$$n_m = 1 - \delta_m - i\beta_m, \quad \delta_m, \beta_m \in \mathbb{R}, \quad (1.3)$$

and the medium occupies the interval

$$I_m = (z_m^-, z_m^+). \quad (1.4)$$

One or both endpoints may be infinite for a terminal medium. A fixed half-open convention must be used in code when an atom lies exactly on a boundary.

orGUI uses the negative-root Renaud normal wavevector

$$k_{z,m} = -k_0 \sqrt{n_m^2 - n_0^2 \cos^2 \alpha}, \quad \operatorname{Re} k_{z,m} \leq 0, \quad \operatorname{Im} k_{z,m} \geq 0. \quad (1.5)$$

The incident and reciprocal exit solutions are calculated in the same frozen array of complex refractive indices and at the same physical boundaries.

Equation (1.5) fixes what “upward” means, and every sign in this document follows from it. Because $\operatorname{Re} k_{z,m} \leq 0$, a wave travelling towards $+z$ — upward, out of the sample and towards the detector — carries $\exp(+ik_{z,m}z)$, while a wave travelling towards $-z$, downward onto the sample, carries $\exp(-ik_{z,m}z)$. This is the opposite pairing from the one a positive-root convention would give, and it is the single most common source of sign errors when comparing with the literature. The whole of Secs. 1.4–1.6 is carried out in Eq. (1.5)’s convention, so no conversion is ever needed.

1.3 Scattering-factor density and the index contrast

For atom a , orGUI uses

$$f_a(Q^2, E) = f_{a,\text{WK}}^0(Q^2) + f'_a(E) + if''_a(E), \quad (1.6)$$

where the Waasmaier–Kirkel representation is evaluated analytically at the generally complex channel value Q^2 . Its zero-transfer value, including the same anomalous terms, is also used to generate the optical profile. Replacing that value by an independent atomic-number approximation would break the reference-density cancellation. Equation (1.6) is the only quantity in this document called f .

Let $\rho_f(\mathbf{r})$ be the complex scattering-factor density in electrons per Angstrom³. To first order in the X-ray susceptibility,

$$n^2(\mathbf{r}) \simeq 1 - \frac{4\pi r_e}{k_0^2} \rho_f(\mathbf{r}), \quad (1.7)$$

with r_e the classical electron radius. The piecewise-constant reference density represented by Eq. (1.3) is therefore

$$\boxed{\bar{\rho}_{f,m} = \frac{k_0^2}{2\pi r_e} (\delta_m + i\beta_m)}. \quad (1.8)$$

Write $n_0^2(z) = \sum_m n_m^2 \mathbf{1}_{I_m}(z)$ for the prepared, laterally uniform reference and

$$\delta n^2(\mathbf{r}) = n^2(\mathbf{r}) - n_0^2(z) = -\frac{4\pi r_e}{k_0^2} \Delta \rho_f(\mathbf{r}), \quad \Delta \rho_f(\mathbf{r}) = \rho_f(\mathbf{r}) - \sum_m \bar{\rho}_{f,m} \mathbf{1}_{I_m}(z), \quad (1.9)$$

for the index contrast that drives all scattering. Equations (1.3) and (1.8) enter the problem only through $n_0^2(z)$, and Sec. 1.6 shows exactly where they survive.

1.4 The Born expansion

At the small scattering angles of interest the vectorial propagation equation reduces to a scalar Helmholtz equation for the component of the field along the analysed polarization (Renaud §5.3.6). Writing E for that component,

$$[\nabla^2 + k_0^2 n^2(\mathbf{r})] E(\mathbf{r}) = 0, \quad n^2(\mathbf{r}) = n_0^2(z) + \delta n^2(\mathbf{r}), \quad (1.10)$$

so that

$$[\nabla^2 + k_0^2 n_0^2(z)] E(\mathbf{r}) = -k_0^2 \delta n^2(\mathbf{r}) E(\mathbf{r}). \quad (1.11)$$

Defining the Green function of the *reference* problem by

$$[\nabla^2 + k_0^2 n_0^2(z)] G(\mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}'), \quad (1.12)$$

Eq. (1.11) is equivalent to the integral equation

$$E(\mathbf{r}) = \mathbf{E}_0(\mathbf{r}) - k_0^2 \int d^3 r' G(\mathbf{r}, \mathbf{r}') \delta n^2(\mathbf{r}') E(\mathbf{r}'), \quad (1.13)$$

whose iteration is the Born series. Retaining the first order, with $\mathbf{E}_0$ and G both those of the *stratified* reference, gives the distorted-wave Born approximation:

$$\boxed{\delta E(\mathbf{r}) = -k_0^2 \int d^3 r' G(\mathbf{r}, \mathbf{r}') \delta n^2(\mathbf{r}') \mathbf{E}_0(\mathbf{r}', \mathbf{k}_i).} \quad (1.14)$$

Reflection, transmission, refraction, and repeated reflection at the planar interfaces are already contained in $\mathbf{E}_0$ and in G ; the perturbation is scattered exactly once.

Sign remark. Renaud's Eqs. (71) and (73) as printed carry $+k_0^2 \delta n^2$ on the right-hand side. With his Eq. (70)'s definition $\delta n^2 = n^2 - n_0^2$, adopted here in Eq. (1.9), the sign is as in Eqs. (1.11) and (1.13). The overall sign of the final result is verified independently against the Born–Fresnel limit in Sec. 1.6.4.

1.5 The Green function of the stratified reference

This section does the work that is usually skipped. Nothing in it is approximate.

1.5.1 Separation into a one-dimensional problem

Because n_0^2 depends only on z , Eq. (1.12) separates under a lateral Fourier transform:

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{(2\pi)^2} \int d^2 q g_q(z, z') e^{-i\mathbf{q} \cdot (\mathbf{r}_{\parallel} - \mathbf{r}'_{\parallel})}, \quad \left[\frac{d^2}{dz^2} + k_0^2 n_0^2(z) - q^2 \right] g_q(z, z') = \delta(z - z'). \quad (1.15)$$

The lateral wavevector $\mathbf{q}$ enters the one-dimensional operator only through the constant $-q^2$; the normal wavevector belonging to $\mathbf{q}$ in medium m is Eq. (1.5)'s $k_{z,m}$.

1.5.2 Wronskian construction

Equation (1.15) is a one-dimensional Green-function problem for an operator with no first-derivative term. Let u_- and u_+ solve the homogeneous equation subject to one physical boundary condition each,

$$u_-(z) : \text{ no wave incoming from below, as } z \rightarrow -\infty, \quad (1.16)$$

$$u_+(z) : \text{ no wave incoming from above, as } z \rightarrow +\infty, \quad (1.17)$$

each fixed up to one multiplicative constant. Then

$$g_q(z, z') = \frac{u_-(z_{<}) u_+(z_{>})}{W}, \quad W = u_- u'_+ - u'_- u_+, \quad (1.18)$$

with $z_{<} = \min(z, z')$ and $z_{>} = \max(z, z')$. The Wronskian is independent of z ,

$$\frac{dW}{dz} = u_- u''_+ - u''_- u_+ = u_- (q^2 - k_0^2 n_0^2) u_+ - (q^2 - k_0^2 n_0^2) u_- u_+ = 0, \quad (1.19)$$

precisely because the operator has no d/dz term, so it may be evaluated wherever it is easiest. The normalisation of $u_{\pm}$ cancels in Eq. (1.18).

1.5.3 What the two solutions are physically

Using the convention established after Eq. (1.5), in vacuum ($m = 0$):

- u_+ is fixed by Eq. (1.17): above the whole structure it is purely upward-going,

$$u_+(z) = e^{+ik_{z,0}z}. \quad (1.20)$$

How it continues downward through the stack is never needed, because the observation point is always in vacuum.

- u_- is fixed by Eq. (1.16): deep in the terminal medium it carries only the transmitted, downward or evanescently decaying wave, which is the physical condition $A_M^- = 0$. That single condition, continued *upward* through the stack, produces in vacuum a superposition of a downward and an upward component; normalising the downward one to unity,

$$u_-(z) = e^{-ik_{z,0}z} + r_0 e^{+ik_{z,0}z} \quad (\text{in vacuum}). \quad (1.21)$$

This is the ordinary Fresnel/Parratt reference field for a plane wave incident from above — incident plus reflected, with only a transmitted wave below — and r_0 is the unperturbed specular reflection amplitude of Eq. (4.1). The condition that *defines* u_- is imposed at $z \rightarrow -\infty$, but the solution it selects is the familiar one driven from above.

Since the exit wavevector $\mathbf{k}_f$ points away from the sample, $-\mathbf{k}_f$ points onto it, and u_- evaluated for that direction is exactly the reciprocal exit field,

$$u_-(z') = \mathbf{E}_0(z', -\mathbf{k}_f). \quad (1.22)$$

This is the content of the reciprocity theorem, obtained here by solving a boundary-value problem rather than by propagating a field to a distant detector. It is why the reciprocal exit field is never complex conjugated: it is a solution of the same homogeneous equation as the incident field, not its adjoint.

1.5.4 The Wronskian, and the Green function

Evaluate W in vacuum from Eqs. (1.20)–(1.21):

$$\begin{aligned} W &= \left[e^{-ik_{z,0}z} + r_0 e^{ik_{z,0}z} \right] \left(ik_{z,0} e^{ik_{z,0}z} \right) - \left[-ik_{z,0} e^{-ik_{z,0}z} + ik_{z,0} r_0 e^{ik_{z,0}z} \right] e^{ik_{z,0}z} \\ &= ik_{z,0} + ik_{z,0} r_0 e^{2ik_{z,0}z} + ik_{z,0} - ik_{z,0} r_0 e^{2ik_{z,0}z} = 2ik_{z,0}. \end{aligned} \quad (1.23)$$

The r_0 terms cancel identically: *the Wronskian does not depend on the stack*, only on the vacuum $k_{z,0}$. This one fact is the origin of every factor $1/k_z$ in this subject, including the one that appears without explanation in the reflectivity literature.

With the observation point above all sources, $z > z'$, so $z_< = z'$, $z_> = z$, and Eqs. (1.18), (1.20), (1.22) and (1.23) give

$$g_q(z, z') = \frac{\mathbf{E}_0(z', -\mathbf{k}_f) e^{ik_{z,0}z}}{2ik_{z,0}} = -\frac{i}{2k_{z,0}} e^{ik_{z,0}z} \mathbf{E}_0(z', -\mathbf{k}_f). \quad (1.24)$$

For a homogeneous reference, $r_0 = 0$ and $u_-(z') = e^{-ik_{z,0}z'}$, so Eq. (1.24) collapses to $g_q = -(i/2k_{z,0})e^{ik_{z,0}|z-z'|}$, whose derivative jump across $z = z'$ is unity; substituting that into Eq. (1.15) reproduces the Weyl angular-spectrum representation of an outgoing spherical wave. The Weyl expansion is therefore not an extra ingredient but the vacuum special case of Eq. (1.24).

1.5.5 Why the far-field route is not available here

The textbook shortcut replaces G by its far-field limit, expanding $|\mathbf{r} - \mathbf{r}'| \simeq r - \mathbf{k}_f \cdot \mathbf{r}'/k_0$ and pulling a single spherical wave $e^{-ik_0 r}/r$ out of the integral. That expansion requires the detector distance R to exceed the Fraunhofer distance of the *whole illuminated footprint*. For a grazing-incidence footprint of lateral size $L \sim 1$ mm to 1 cm at $\lambda \sim 1$ Å, the neglected quadratic phase is

$$\delta(\text{phase}) \sim k_0 \frac{L^2}{R} \sim 10^4\text{--}10^7 \gg 1, \quad (1.25)$$

so the far-field form of G is never valid on the specular rod. Equation (1.24) avoids the issue entirely: it is exact at any R .

The polarization factor is a separate matter. The same far-field substitution is conventionally used to fix the exit polarization vector, but there it enters undifferentiated and is *not* multiplied by k_0 , so its error is only $\sim L/R \sim 10^{-2}$ rather than Eq. (1.25). Fixing the polarization basis at the nominal $\mathbf{k}_f$, as Eq. (2.7) does, therefore remains accurate even where the phase treatment above is mandatory.

1.6 Lattice expansion and the lateral collapse

1.6.1 The index contrast along the surface lattice

Parallel to the surface, $n^2(\mathbf{r})$ is periodic on the two-dimensional lattice of the reference lateral cell, of area A_{ref} , while $n_0^2(z)$ is laterally uniform. Their difference δn^2 is therefore two-dimensional-periodic and has the Fourier series

$$\delta n^2(\mathbf{r}_{\parallel}, z) = \sum_{\mathbf{h}} \delta n_{\mathbf{h}}^2(z) e^{-i\mathbf{h} \cdot \mathbf{r}_{\parallel}}, \quad \delta n_{\mathbf{h}}^2(z) = \frac{1}{A_{\text{ref}}} \int_{\text{cell}} d^2 r_{\parallel} \delta n^2(\mathbf{r}_{\parallel}, z) e^{i\mathbf{h} \cdot \mathbf{r}_{\parallel}}, \quad (1.26)$$

where $\mathbf{h}$ runs over the reciprocal lattice of that cell, i.e. over the crystal truncation rods (h, k) . Because $n_0^2(z)$ is laterally uniform it contributes to Eq. (1.26) at $\mathbf{h} = \mathbf{0}$ and nowhere else:

$$\delta n_{\mathbf{h}}^2(z) = \frac{1}{A_{\text{ref}}} \int_{\text{cell}} d^2 r_{\parallel} n^2(\mathbf{r}_{\parallel}, z) e^{i\mathbf{h} \cdot \mathbf{r}_{\parallel}}, \quad \mathbf{h} \neq \mathbf{0}, \quad (1.27)$$

$$\delta n_{\mathbf{0}}^2(z) = \frac{1}{A_{\text{ref}}} \int_{\text{cell}} d^2 r_{\parallel} n^2(\mathbf{r}_{\parallel}, z) - n_0^2(z), \quad \mathbf{h} = \mathbf{0}. \quad (1.28)$$

Equations (1.27)–(1.28) are the entire content of the specular/non-specular distinction, and they are the reason this document needs only one matrix element. The δ_m, β_m of the prepared profile are not something subtracted afterwards from an atomic sum: they are part of the $\mathbf{h} = \mathbf{0}$ Fourier coefficient, and part of nothing else.

1.6.2 Collapse

Insert Eqs. (1.15), (1.24) and (1.26) into Eq. (1.14). Stratification lets the lateral phase of the incident field be factored out, $\mathbf{E}_0(\mathbf{r}', \mathbf{k}_i) = e^{-i\mathbf{k}_{i\parallel} \cdot \mathbf{r}'_{\parallel}} \mathbf{E}_0(z', \mathbf{k}_i)$, so all $\mathbf{r}'_{\parallel}$ dependence sits in three exponentials and the lateral integral is elementary:

$$\int d^2 r'_{\parallel} e^{i\mathbf{q} \cdot \mathbf{r}'_{\parallel}} e^{-i\mathbf{h} \cdot \mathbf{r}'_{\parallel}} e^{-i\mathbf{k}_{i\parallel} \cdot \mathbf{r}'_{\parallel}} = (2\pi)^2 \delta^{(2)}(\mathbf{q} - \mathbf{k}_{i\parallel} - \mathbf{h}). \quad (1.29)$$

The delta function is the Laue condition: for each rod exactly one angular-spectrum mode survives,

$$\mathbf{q} = \mathbf{k}_{i\parallel} + \mathbf{h} \equiv \mathbf{k}_{f\parallel}, \quad \text{that is} \quad \mathbf{Q}_{\parallel} = \mathbf{h}, \quad (1.30)$$

at any incidence angle, any exit angle, and any azimuth. The $1/r$ spherical falloff has cancelled against the density of angular-spectrum modes: what emerges is a genuine, non-decaying plane wave per rod.

1.6.3 The matrix element and the rod amplitude

Collecting the constants of Eqs. (1.14), (1.15), (1.24) and (1.29),

$$-k_0^2 \cdot \frac{1}{(2\pi)^2} \cdot \left(-\frac{i}{2k_{z,0,f}} \right) \cdot (2\pi)^2 = +\frac{ik_0^2}{2k_{z,0,f}}, \quad (1.31)$$

and converting from δn^2 to $\Delta\rho_f$ with Eq. (1.9), the k_0^2 cancels exactly. Defining the matrix element of rod $\mathbf{h}$, normalised to one reference lateral cell,

$$F_{\mathbf{h}} = \int dz \int_{\text{cell}} d^2r_{\parallel} \Delta\rho_f(\mathbf{r}_{\parallel}, z) e^{i\mathbf{h} \cdot \mathbf{r}_{\parallel}} \mathbf{E}_0(z, -\mathbf{k}_f) \cdot \mathbf{E}_0(z, \mathbf{k}_i), \quad (1.32)$$

the scattered field on rod $\mathbf{h}$ is the plane wave

$$r_{\mathbf{h}} = -\frac{2\pi i r_e}{A_{\text{ref}} k_{z,0,f}} F_{\mathbf{h}}. \quad (1.33)$$

Equation (1.32) is the only matrix element in this document. It is used unchanged for every rod:

- for $\mathbf{h} \neq \mathbf{0}$ its integrand contains no reference term, by Eq. (1.27);
- for $\mathbf{h} = \mathbf{0}$ its integrand contains the constant $\delta_m + i\beta_m$ density of every prepared slab, by Eq. (1.28), through $\Delta\rho_f$'s $-\sum_m \bar{\rho}_{f,m} \mathbf{1}_{I_m}$ term.

Chapter 2 evaluates Eq. (1.32) for atoms, producing the four coherent channels. Chapter 3 evaluates the reference part of $F_{\mathbf{0}}$ and partitions it among generated records. The reciprocal exit field in Eq. (1.32) is not complex conjugated; conjugation enters only after the coherent amplitude has been formed.

1.6.4 Born–Fresnel check

Take a uniform half-space $z < 0$ of density ρ_f against a vacuum reference, so that only $\mathbf{h} = \mathbf{0}$ contributes, the field product reduces to $e^{iQ_z z}$, and $\int_{-\infty}^0 e^{iQ_z z} dz = 1/(iQ_z)$. Then $F_{\mathbf{0}} = A_{\text{ref}} \rho_f / (iQ_z)$ and, with $Q_z = -2k_{z,0} > 0$ from Eq. (2.4),

$$r_{\mathbf{0}} = -\frac{2\pi i r_e}{A_{\text{ref}} k_{z,0}} \cdot \frac{A_{\text{ref}} \rho_f}{iQ_z} = -\frac{2\pi r_e \rho_f}{k_{z,0} Q_z} = +\frac{4\pi r_e \rho_f}{Q_z^2} > 0, \quad (1.34)$$

which is the known kinematic reflectivity amplitude, with the positive sign of the exact Fresnel expansion $r = (k_{z,0} - k_{z,1}) / (k_{z,0} + k_{z,1})$ for $n < 1$. Both the sign and the constant in Eq. (1.33) are therefore fixed independently of any external sign convention.

Chapter 2

Evaluating $F_{\mathbf{h}}$: atoms and the four coherent channels

This chapter evaluates the matrix element $F_{\mathbf{h}}$ of Eq. (1.32) for the atomic part of the contrast. The reference part, which exists only at $\mathbf{h} = \mathbf{0}$, is evaluated in Chapter 3. Both are contributions to the same $F_{\mathbf{h}}$; they are never separate matrix elements.

2.1 Where the four channels come from

Section 1.5 showed that the reciprocal exit field entering Eq. (1.32) is u_- , the solution of the homogeneous stratified problem selected by the boundary condition below the stack. Inside prepared medium m that solution has two branches, and so does the incident field:

$$\mathbf{E}_0(z', -\mathbf{k}_f) = A_{m,f}^+ e^{-ik_{z,m,f}z'} + A_{m,f}^- e^{+ik_{z,m,f}z'}, \quad \mathbf{E}_0(z', \mathbf{k}_i) = A_{m,i}^+ e^{-ik_{z,m,i}z'} + A_{m,i}^- e^{+ik_{z,m,i}z'}. \quad (2.1)$$

Their product in Eq. (1.32) therefore contains four terms, $A_{m,i}^{\sigma_i} A_{m,f}^{\sigma_f}$. The four channels are not an additional model: they are the two branches of each reference field, and nothing more needs to be assumed to justify them. Let

$$\sigma_i, \sigma_f \in \{+1, -1\}, \quad \nu = (\sigma_i, \sigma_f), \quad (2.2)$$

with the fixed channel order

$$++, \quad +-, \quad -+, \quad --. \quad (2.3)$$

2.2 Channel wavevectors and coefficients

For medium m , channel ν has

$$\mathbf{Q}_m^\nu = (\mathbf{Q}_\parallel, Q_{z,m}^\nu), \quad Q_{z,m}^\nu = -(\sigma_i k_{z,m,i} + \sigma_f k_{z,m,f}), \quad (2.4)$$

with $\mathbf{Q}_\parallel = \mathbf{h}$ by the Laue condition, Eq. (1.30), and

$$(Q_m^\nu)^2 = |\mathbf{h}|^2 + (Q_{z,m}^\nu)^2. \quad (2.5)$$

Neither $Q_{z,m}^\nu$ nor $(Q_m^\nu)^2$ is generally real in an absorbing medium. Because Eq. (1.5) makes $\text{Re } k_{z,m} \leq 0$, Eq. (2.4) gives $Q_z > 0$ for the $++$ channel, as it must.

Let $A_{m,i}^{\sigma_i}$ and $A_{m,f}^{\sigma_f}$ be the local Renaud electric-field amplitudes of Eq. (2.1), referenced by the field solver to $z_{m,i}^{\text{ref}}$ and $z_{m,f}^{\text{ref}}$. The complete coefficient multiplying a Fourier factor $\exp(i\mathbf{Q}_m^\nu \cdot \mathbf{r})$ is

$$\begin{aligned} \mathcal{C}_{m,\nu} &= A_{m,i}^{\sigma_i} A_{m,f}^{\sigma_f} P_{m,\nu} \\ &\times \exp \left\{ i \left[\sigma_i k_{z,m,i} z_{m,i}^{\text{ref}} + \sigma_f k_{z,m,f} z_{m,f}^{\text{ref}} \right] \right\}. \end{aligned} \quad (2.6)$$

The reference phase in Eq. (2.6) is not optional: the stored $A^\pm$ are local amplitudes, not amplitudes referenced to the global crystallographic origin.

For the right-handed Renaud polarization basis used by orGUI, the bilinear polarization contractions are

$$\begin{aligned} P_{ss}^{\sigma_i \sigma_f} &= \cos \varphi, \\ P_{sp}^{\sigma_i \sigma_f} &= -\sigma_f \sin \alpha_f \sin \varphi, \\ P_{ps}^{\sigma_i \sigma_f} &= -\sigma_i \sin \alpha_i \sin \varphi, \\ P_{pp}^{\sigma_i \sigma_f} &= \sin \alpha_i \sin \alpha_f \left[\frac{k_0^2 \cos \alpha_i \cos \alpha_f}{k_{z,m,i} k_{z,m,f}} - \sigma_i \sigma_f \cos \varphi \right]. \end{aligned} \quad (2.7)$$

These are evaluated at the nominal $\mathbf{k}_i$ and $\mathbf{k}_f$; by Sec. 1.5.5 that remains accurate to $\sim L/R$ even where a far-field treatment of the *phase* would fail badly.

The solver's upward internal tangential coefficient for p polarization has the opposite sign from the public Renaud amplitude. Thus

$$A_{p,m}^+ = T_{p,m}^+, \quad A_{p,m}^- = -T_{p,m}^-. \quad (2.8)$$

Implementations should use the solver's analytic A^σ/k_z limits in the pp expression rather than divide numerically at a critical angle.

2.3 One generated UnitCell record

The surface model is flattened into generated unit-cell records u . Before flattening, every source **UnitCell** is split into the same ordered structural-layer views used to build the optical profile. A record therefore contains one such layer cell, a crystal-model weight w_u , its coherent domain transforms d , and possibly signed domain occupancies p_{ud} . A one-layer source cell is already a single record. Let A_u be the source cell's lateral area and A_{ref} the crystal-wide reference area of Eq. (1.26). The required area conversion is

$$s_u = \frac{A_{\text{ref}}}{A_u}. \quad (2.9)$$

Current Film and interface domain transforms preserve lateral area. If a future model permits in-plane strain, A_u in Eq. (2.9) must be replaced by the transformed projected area, including its Jacobian.

After stacking and the coherent-domain transform, atom a of record u and domain d is at the physical Cartesian position $\mathbf{r}_{uad}$. The atom is assigned to the unique prepared optical interval

$$m(u, a, d) \quad \text{such that} \quad z_{uad} \in I_{m(u,a,d)}. \quad (2.10)$$

A generated unit cell that crosses an optical boundary must therefore be split by atom. Assigning one medium to the entire cell would use the wrong $A^\pm$, k_z , polarization contraction, and complex form-factor argument for some of its atoms.

For channel ν in medium m , define the atomic contribution

$$\mathcal{A}_{u,m}^\nu = s_u w_u \mathcal{C}_{m,\nu} \sum_d p_{ud} \sum_{\substack{a \in u \\ m(u,a,d)=m}} o_a f_a((Q_m^\nu)^2, E) \exp[-W_a(Q_m^\nu)] \exp(i\mathbf{Q}_m^\nu \cdot \mathbf{r}_{uad}). \quad (2.11)$$

Here o_a is the atomic occupancy. For the currently implemented in-plane and out-of-plane Debye–Waller parameters,

$$W_a(\mathbf{Q}_m^\nu) = \frac{B_{a,\parallel} |\mathbf{h}|^2 + B_{a,\perp} (Q_{z,m}^\nu)^2}{16\pi^2}. \quad (2.12)$$

Equations (1.6), (2.5), and (2.12) are evaluated at complex arguments without taking an absolute value or a real part.

2.4 The atomic contribution to $F_{\mathbf{h}}$

The atomic part of record u 's contribution to Eq. (1.32) is

$$F_{\mathbf{h},u}^{\text{at}} = \sum_m \sum_{\nu \in \{++,+,-,-,-\}} \mathcal{A}_{u,m}^{\nu}, \quad (2.13)$$

and for $\mathbf{h} \neq \mathbf{0}$ this is the whole of it,

$$F_{\mathbf{h}} = \sum_u F_{\mathbf{h},u}^{\text{at}}, \quad \mathbf{h} \neq \mathbf{0}, \quad (2.14)$$

because Eq. (1.27) carries no reference term. The specular coefficient $F_{\mathbf{0}}$ carries one additional contribution and is completed in Chapter 3.

The four terms in Eq. (2.13) are amplitudes and must be added before an intensity is formed. The incorrect expression $\sum_{\nu} |\mathcal{A}_{u,m}^{\nu}|^2$ discards the standing-wave interference between incident and exit branches that Eq. (2.1) makes explicit.

2.5 Useful limits

In a terminal substrate, the boundary condition that defines u_- in Eq. (1.16) gives

$$A_{M,i}^- = A_{M,f}^- = 0, \quad (2.15)$$

so only the $++$ channel survives. In the high-angle Born limit of a finite cell in vacuum, $A_i^+ = A_f^+ = 1$ and all reflected branches vanish; the same reduction occurs. These are limiting cases of the four-channel expression, not reasons to omit channels in finite graded layers.

If the cell is wholly contained in one medium, Eq. (2.11) reduces to four ordinary structure-factor evaluations at the four complex values $Q_{z,m}^{\nu}$. If it crosses a boundary, there is no single structure factor that can be multiplied by one set of channel coefficients; the atom-medium partition in Eq. (2.11) is essential.

Chapter 3

The specular coefficient F_0

3.1 What is different at $\mathbf{h} = \mathbf{0}$

Everything that distinguishes the specular rod was already established in Eqs. (1.27)–(1.28): the prepared reference $n_0^2(z)$ is laterally uniform, so it appears in the $\mathbf{h} = \mathbf{0}$ Fourier coefficient of δn^2 and in no other. Equivalently, in the language of Eq. (1.29), a laterally uniform density has all of its lateral Fourier weight at $\mathbf{q} - \mathbf{k}_{i\parallel} = \mathbf{0}$, and a genuine rod $\mathbf{h} \neq \mathbf{0}$ never sits there.

Consequently F_0 , and only F_0 , receives a second contribution besides the atomic sum of Eq. (2.13):

$$F_0 = \sum_u \left(F_{0,u}^{\text{at}} + F_{0,u}^{\text{ref}} \right), \quad F_{\mathbf{h}} = \sum_u F_{\mathbf{h},u}^{\text{at}} \quad (\mathbf{h} \neq \mathbf{0}). \quad (3.1)$$

This is one matrix element with two kinds of contribution, not two matrix elements. In particular nothing is “subtracted afterwards”: the sign is already carried by $\Delta\rho_f$ in Eq. (1.9).

It is worth stating what would go wrong if the reference contribution were instead applied cell by cell off-specular. Doing so would replace the exact lateral delta function of Eq. (1.29) by a finite-cell shape transform and manufacture diffuse scattering that the reference profile does not produce. Instrumental resolution may mix genuine diffuse intensity into a detector pixel on the specular rod, but that belongs to a later resolution convolution, not to an interpolation of Eq. (3.1) near $\mathbf{h} = \mathbf{0}$.

3.2 Integral of one constant optical slab

Within prepared medium m , both $\bar{\rho}_{f,m}$ and every channel coefficient are constant; only the propagation phase varies with z . Define the stable depth integral

$$\Phi(q; a, b) = \int_a^b e^{iqz} dz = \begin{cases} \frac{e^{iqb} - e^{iqua}}{iq}, & q \neq 0, \\ b - a, & q = 0. \end{cases} \quad (3.2)$$

For numerical work, an equivalent midpoint form is preferable:

$$\Phi(q; a, b) = (b - a) e^{iq(a+b)/2} \text{sinc}\left(\frac{q(b-a)}{2}\right), \quad \text{sinc } x = \frac{\sin x}{x}, \quad (3.3)$$

or the numerator of Eq. (3.2) may be evaluated with a complex `expm1`. For a lower terminal half-space, with the physical decaying branch,

$$\Phi(q; -\infty, b) = \frac{e^{iqb}}{iq}. \quad (3.4)$$

Carrying the $-\sum_m \bar{\rho}_{f,m} \mathbf{1}_{I_m}$ term of Eq. (1.9) through Eq. (1.32), the reference contribution to F_0 at profile level is

$$\sum_u F_{0,u}^{\text{ref}} = -A_{\text{ref}} \sum_m \bar{\rho}_{f,m} \sum_{\nu \in \{++,+,-,-,-\}} \mathcal{C}_{m,\nu} \Phi(Q_{z,m}^\nu; z_m^-, z_m^+). \quad (3.5)$$

Finite graded layers generally retain all four channels in Eq. (3.5); only the terminal substrate reduces to $++$, by Eq. (2.15).

3.3 A well-defined per-record partition

Only the total reference profile is physical. A decomposition of that profile among overlapping model objects is bookkeeping and is not unique unless a rule is fixed. orGUI can make the decomposition deterministic by retaining the forward-density contribution of every generated record while building the profile.

Let $\bar{\rho}_{f,u,m}$ denote record u 's share of the constant reference density in prepared medium m . The required conservation rule is

$$\sum_u \bar{\rho}_{f,u,m} = \bar{\rho}_{f,m} \quad \text{for every prepared medium } m. \quad (3.6)$$

Before optical-profile simplification, a generated slab of physical thickness t_{ud} contributes naturally

$$\bar{\rho}_{f,ud} = \frac{w_u p_{ud}}{A_u t_{ud}} \sum_{a \in u} o_a f_a(0, E). \quad (3.7)$$

Normal strain changes t_{ud} and hence the volume density, but preserves the integrated areal content. Signed occupancies in interface and etching models remain signed in Eq. (3.7).

Coincident profile contributions are added. If adjacent optical media are merged, first determine the merged boundaries from the total profile, then thickness-average every record's density over those same boundaries. This is a linear operation once the grouping is fixed and preserves Eq. (3.6). Reconstructing shares independently with different merge decisions would not preserve the prepared reference.

With Eq. (3.6) fixed, record u 's reference contribution is

$$F_{0,u}^{\text{ref}} = -A_{\text{ref}} \sum_m \bar{\rho}_{f,u,m} \sum_\nu \mathcal{C}_{m,\nu} \Phi(Q_{z,m}^\nu; z_m^-, z_m^+), \quad (3.8)$$

so that record u 's complete share of the specular matrix element is

$$F_{0,u} = \sum_m \sum_\nu \mathcal{A}_{u,m}^\nu - A_{\text{ref}} \sum_m \bar{\rho}_{f,u,m} \sum_\nu \mathcal{C}_{m,\nu} \Phi(Q_{z,m}^\nu; z_m^-, z_m^+). \quad (3.9)$$

Summing Eq. (3.9) over u and applying Eq. (3.6) recovers Eq. (3.5) and hence the profile-level F_0 of Eq. (3.1):

$$F_0 = \sum_u F_{0,u} = \sum_u F_{0,u}^{\text{at}} - A_{\text{ref}} \sum_m \bar{\rho}_{f,m} \sum_\nu \mathcal{C}_{m,\nu} \Phi(Q_{z,m}^\nu; z_m^-, z_m^+). \quad (3.10)$$

If callers do not need an attribution by object, Eq. (3.10) is the preferred calculation: accumulate all atomic terms, evaluate the prepared reference once, and add once.

3.4 Reduction to the implemented terminal bulk formula

Let the top terminal bulk cell occupy $b - c \leq z < b$, with repeat vector $-\mathbf{c}$ into the substrate and $c = c_z > 0$. Only the $++$ channel survives, by Eq. (2.15). If F_{uc} denotes the ordinary unit-cell structure factor including the global phase of the top cell, the semi-infinite atomic contribution to $F_{\mathbf{0}}$ is

$$F_{\mathbf{0},\text{bulk}}^{\text{at}} = \mathcal{C}_{M,++} \frac{F_{\text{uc}}}{1 - \exp(-i\mathbf{Q}_M^{++} \cdot \mathbf{c})}, \quad (3.11)$$

and, for a surface-normal repeat, the mean-density contribution is

$$F_{\mathbf{0},\text{bulk}}^{\text{ref}} = -\mathcal{C}_{M,++} A_{\text{ref}} \bar{\rho}_{f,M} \frac{e^{iQ_{z,M}^{++} b}}{iQ_{z,M}^{++}}. \quad (3.12)$$

Their sum is the previously implemented bulk result. The four-channel finite-layer formula therefore extends the existing kernel without changing the terminal half-space convention.

Equation (3.12) also shows why the reference contribution cannot be dropped: it is precisely the $Q_z \rightarrow 0$ divergence of the kinematic lattice sum in Eq. (3.11), that is, the $\mathbf{h} = \mathbf{0}$ component that the prepared optical profile already describes and that r_0 in Eq. (4.1) already carries. Retaining both without Eq. (3.10) would count it twice.

Chapter 4

Mapping to orGUI models and validation

4.1 Flattened atomic representation

The mathematical kernel should consume a flattened list of generated atomic records rather than special-case each high-level model. After stacking, the records have the following meanings.

Model	Records supplied to the F_h sum
UnitCell	Ordered structural-layer views, atoms, source lateral area, internal coherent-domain transforms and occupancies, and the crystal-wide component weight.
Film	The finite cyclic layer cells generated by <code>createLayers</code> . Their domain translations place them at the physical stacked heights.
EpitaxyInterface	Positive upper-material probability records; positive lower-material records at their possibly strain-coupled positions; and negative lower-material records that remove the sharp, unstrained bulk already present in the terminal lattice sum.
PoissonSurface	Positive records for grown material and negative records for material removed from the sharp Film termination. Termination corrections remain coherent with the Film amplitude.

Negative occupancies are amplitudes, not negative intensities. They must pass unchanged through Eqs. (2.11) and (3.7). Likewise, alternative crystal domains are applied as coherent amplitudes within each configured domain; intensities are combined only at the model level where the existing API explicitly defines an incoherent mixture.

Water and other continuous-density models do not have an atomic UnitCell sum and are outside this chapter. They may use the same slab integral in Eq. (3.5), but their actual-density integral must be derived from their own continuous profile.

4.2 Preparation and evaluation boundary

The prepared object must freeze all quantities that define the reference problem:

- the physical optical boundaries and the final, possibly simplified, values δ_m, β_m ;
- incident and reciprocal exit k_z , $A^\pm$, $A^\pm/k_z$, and reference planes for every medium;
- incident and exit polarization, external geometry, and the rod $\mathbf{h} = \mathbf{Q}_\parallel$;

- the reference-area convention and the per-record reference shares satisfying Eq. (3.6).

Atomic positions, occupancies, Debye–Waller parameters, and species may be reevaluated against a frozen reference if that is the intended fit operation. Changing lattice transforms, stacking geometry, photon energy, optical boundaries, profile simplification, or the reference area invalidates the preparation. Rebuilding only the atomic list while silently changing the optical profile would combine two different reference problems.

For each scattering point, evaluation is naturally ordered as follows:

1. identify the rod from the Laue condition, Eq. (1.30), and calculate $\mathbf{Q}_m^\nu$, $P_{m,\nu}$, and $\mathcal{C}_{m,\nu}$ for all media and all four channels;
2. assign every transformed atom to its physical optical medium;
3. accumulate Eq. (2.11) coherently over atoms, domains, media, and channels, giving $F_{\mathbf{h},u}^{\text{at}}$;
4. if $\mathbf{h} = \mathbf{0}$, add the constant-profile contribution of Eq. (3.5), completing $F_{\mathbf{0}}$; otherwise $F_{\mathbf{h}}$ is already complete;
5. convert to a rod amplitude with Eq. (1.33), and only then form an intensity or combine with the prepared unperturbed reflection amplitude.

4.3 Observables

Equation (1.33) already gives the amplitude of the emergent plane wave on rod $\mathbf{h}$,

$$r_{\mathbf{h}} = -\frac{2\pi i r_e}{A_{\text{ref}} k_{z,0,f}} F_{\mathbf{h}},$$

normalised to one reference lateral cell, for every rod and every scattering geometry. Nothing further needs to be assumed about the measurement geometry: fixed α_i , fixed α_f , $\alpha_i = \alpha_f$, coplanar or not, the scan mode only decides which (α_i, α_f) pair — and hence which $k_{z,m}$ and which $A^\pm$ — are evaluated at each measured point.

On a genuine crystal truncation rod, $\mathbf{h} \neq \mathbf{0}$, there is no unperturbed beam in that direction and the observable follows from $r_{\mathbf{h}}$ alone. On the specular rod the scattered field is combined coherently with the prepared plane-wave reflection amplitude,

$$r_{\text{total}} = r_0 + r_{\mathbf{0}}, \quad R_{\text{coh}} = |r_{\text{total}}|^2. \quad (4.1)$$

The r_0 in Eq. (4.1) must be the same r_0 that appears in Eq. (1.21), that is, it must come from the same prepared incident field as the $A_{m,i}^\pm$ used in the matrix element. For unpolarized radiation, calculate s and p independently and average their intensities; do not add their amplitudes.

4.4 Returned quantities

The kernel returns one matrix element and everything else is derived from it by the prefactor of Eq. (1.33). Implementations write that prefactor with the positive vacuum exit wavevector $\kappa_f = k_0 \sin \alpha_f = -k_{z,0,f}$, so the sign appears reversed relative to Eq. (1.33); the two are identical.

Quantity	Definition	Units
$F_{\mathbf{h}}$	Eq. (1.32)	electrons per A_{ref}
$r_{\mathbf{h}}$	$2\pi i r_e F_{\mathbf{h}} / (A_{\text{ref}} \kappa_f)$	dimensionless
r_0	Fresnel amplitude of the prepared reference, Eq. (1.21)	dimensionless
r	$r_0 + r_{\mathbf{h}}$, with $r_0 = 0$ for $\mathbf{h} \neq \mathbf{0}$	dimensionless
F_{eff}	$A_{\text{ref}} \kappa_f r / (2\pi i r_e)$	electrons per A_{ref}
R	$ r ^2$	dimensionless

Three points are worth stating explicitly, because each has caused confusion.

Where r_e belongs. The classical electron radius enters once, inside the prefactor. The separate convention, in which a bare matrix element is turned into a cross-section kernel by $r_e^2 |F_{\mathbf{h}}|^2$, applies to $F_{\mathbf{h}}$ and not to r . The two must never be mixed.

What F_{eff} is. Inverting the prefactor on the *total* amplitude gives the structure factor that a kinematic analysis would infer from this reflectivity. It is a DWBA output; it is not the structure factor of a separate kinematical model, which the tutorial material writes F_{kin} . The two are compared against each other, never substituted for one another. It is not linear in the model density, because r_0 is zeroth order and exact in the reference to all orders, so it is not the DWBA structure factor of the sample; that is $F_{\mathbf{h}}$. It vanishes as $\alpha_f \rightarrow 0$ and is meaningful only well above the critical angle. Off-specular, where $r_0 = 0$, it degenerates to $F_{\mathbf{h}}$. Its value is the large- Q_z limit, where $r_0 \rightarrow 4\pi r_e \bar{\rho}_f / Q_z^2$ and $F_{\text{eff}} \rightarrow F_{\mathbf{h}} + A_{\text{ref}} \bar{\rho}_f / (iQ_z)$, that is, the full kinematic structure factor of the actual density — exactly the quantity a kinematically reduced data set contains.

Why $R = |r|^2$ is the reflectivity to use. On the specular rod at small angles the problem is genuinely one-dimensional, so the scalar reduction of Eq. (1.10) involves no approximation and the s and p channels become degenerate; and r_0 carries the exact Fresnel and refraction response of the prepared reference to all orders. The only error is therefore second order in the density contrast, and it shrinks with decreasing angle: below the critical angle the exact reflectivity is unity and $|r_0 + r_{\mathbf{h}}|^2$ reproduces it. A reference whose density is wrong by 10% gives a relative error near 10^{-4} at one tenth of the critical angle, rising to a few percent near α_c , and falling quadratically as the contrast error is reduced. Squaring the amplitude rather than expanding it also keeps R non-negative. The strictly linearised variant, $|r_0|^2 + 2 \text{Re}(r_0^* r_{\mathbf{h}})$, is useful only as a diagnostic: a large difference between the two means $|r_{\mathbf{h}}|$ is not small against $|r_0|$ and the first-order treatment is being pushed beyond its regime.

4.5 Regression invariants

The following checks are required before the atomistic extension is considered scientifically usable.

1. **Four-channel coherence.** A finite layer with nonzero $A_i^{\pm}$ and $A_f^{\pm}$ equals the direct atom-by-atom evaluation of Eq. (1.32); changing to a sum of channel intensities must fail the test.
2. **Born limit.** Setting $A_i^+ = A_f^+ = 1$ and the other branches to zero recovers the conventional UnitCell structure factor and the standard polarization contraction.
3. **Born–Fresnel amplitude.** A uniform half-space against a vacuum reference reproduces Eq. (1.34), including its sign. This pins the prefactor of Eq. (1.33) independently of any external convention.
4. **Terminal limit.** A finite stack made progressively thicker converges to Eqs. (3.11) and (3.12) away from exact Bragg poles.
5. **Identical actual and reference density.** Replacing every optical slab by its exactly matching atomic forward density gives $F_0 = 0$ within the atomic-truncation accuracy.
6. **Off-specular invariance.** Adding the reference-density bookkeeping does not change any $F_{\mathbf{h}}$ with $\mathbf{h} \neq \mathbf{0}$, as required by Eq. (1.27).
7. **Partition invariance.** Summing per-record results from Eq. (3.9) equals the profile-level Eq. (3.10), including after profile coalescing and simplification.

8. **Area invariance.** Two commensurate descriptions with different source lateral cells give the same amplitude per A_{ref} after Eq. (2.9) is applied to both atomic and continuum terms.
9. **Signed-model identity.** At zero width, zero growth, or zero etching, the positive and negative interface/surface records cancel in both contributions to F_h .
10. **Boundary split.** Moving an atom across an optical boundary changes its channel coefficient and Q_z exactly once; a cell straddling a boundary agrees with the same atom list represented as two records.
11. **Low- Q_z stability.** Equation (3.2) approaches the slab thickness without catastrophic cancellation.
12. **Reciprocity.** Exchanging source and detector geometry obeys the reciprocal-field symmetry of Eq. (1.22) without conjugating the exit field.
13. **Wronskian normalisation.** For any prepared stack, the Wronskian of Eq. (1.23) evaluates to $2ik_{z,0}$ independently of the layer structure. A solver whose $A^\pm$ violate this has an internal normalisation error that Eq. (1.33) would silently absorb.
14. **No duplicate attenuation.** Absorption carried by complex Q_z is not also applied as an empirical depth factor.

4.6 References

The electric-field, signed- k_z , reciprocal-exit-field, and four-channel semantics follow G. Renaud, R. Lazzari, and F. Leroy, “Probing surface and interface morphology with grazing incidence small angle X-ray scattering,” *Surface Science Reports* **64** (2009), 255–380, especially Sections 5.2–5.4, Eq. (91), and the four-event finite-layer examples in Section 6.2. The present note retains their electromagnetic derivation but does not adopt the scalar wavefunction notation used for the optional quantum- mechanical analogy.

Renaud’s Eq. (100) quotes the one-dimensional stratified Green function, with its factor $1/k_z$, without derivation, citing P. M. Morse and H. Feshbach, *Methods of Theoretical Physics* (McGraw–Hill, 1953) for the general construction, together with X. L. Zhou and S. H. Chen, *Physics Reports* **257** (1995), 223–348; A. V. Andreev, A. G. Michette and A. Renwick, *Journal of Modern Optics* **35** (1988), 1667; V. F. Sears, *Physical Review B* **48** (1993), 17477; I. D. Feranchuk, *Physical Review B* **52** (1995), 9214; and A. A. Maradudin and D. L. Mills, *Physical Review B* **11** (1975), 1392. Section 1.5 carries out that construction explicitly in orGUI’s own conventions, so the present document does not depend on any of them.

The terminal crystal-truncation result and the atomic-density truncation caveat are cross-checked against V. M. Kaganer, “Crystal truncation rods in kinematical and dynamical x-ray diffraction theories,” *Physical Review B* **75** (2007), 245425. The perturbation as a difference from a planar reference follows the same physical construction used by G. H. Vineyard, *Physical Review B* **26** (1982), 4146–4159, and by S. K. Sinha *et al.*, *Physical Review B* **38** (1988), 2297–2311.

The companion file `dwba_literature_crosscheck.md` records the broader comparison with the surface-DWBA and reflectivity literature.